

Digital Logic II

Michael C. Hackett

Associate Professor, Computer Science



Lecture Topics

- Combinational Circuits
 - Sign Extend
 - Encoders
 - Decoders
 - Multiplexers
 - Demultiplexers

Combinational Circuits

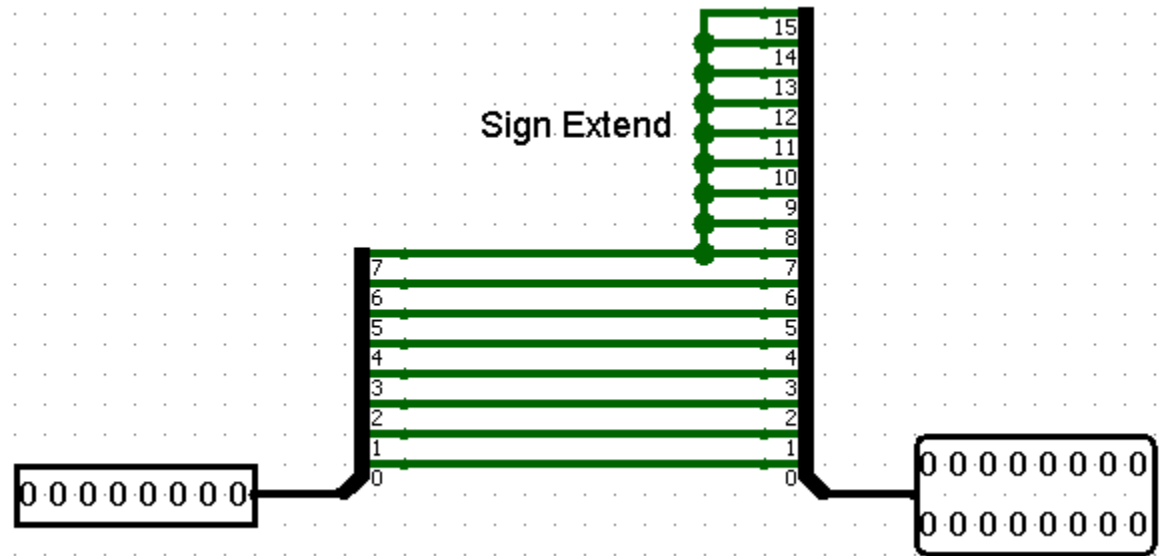
- We've seen the use of logic gates to implement a Boolean function
- With logic gates, we can build **combinational logic circuits** that can perform complex operations
 - They perform Boolean algebra on the circuit's input
- Combinational logic circuits are used in the construction of practical, digital computer circuits (including a computer's CPU)

Sign Extend

- A sign extend is a combinational logic circuit that preserves the sign bit of a number on a bus when the bus's width is increased.
- For example, we'll say two 8-bit (two's complement) binary numbers 11001100 (negative) and 01010101 (positive) are sent on an 8-bit bus
- The bus's width is extended to 16-bits
 - A sign extend preserves the sign bit when this happens

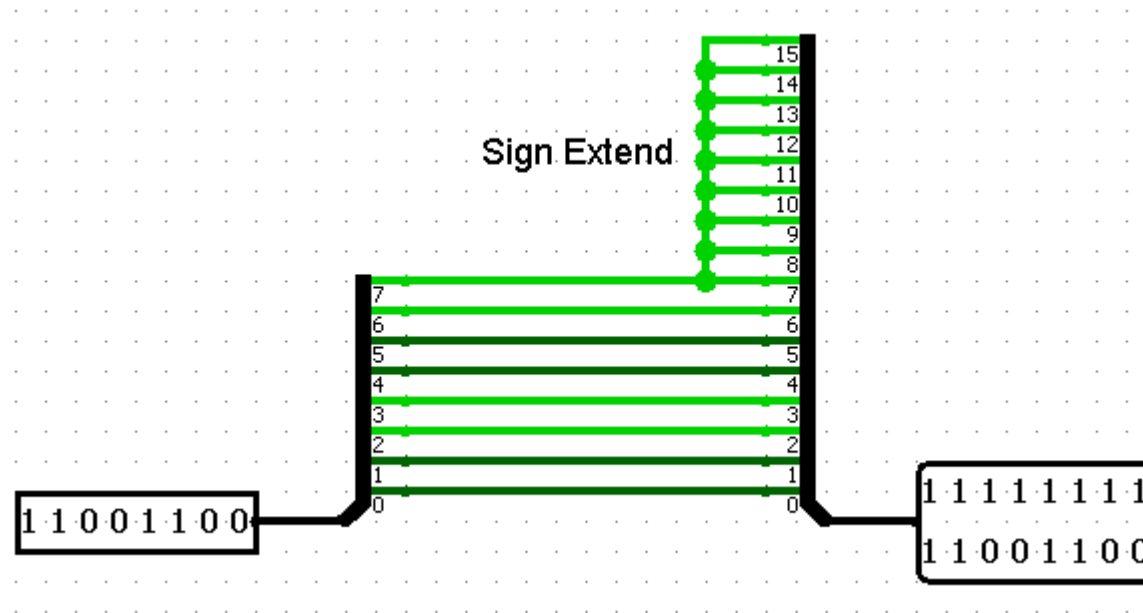
Sign Extend

Input Bus (8-bits)	Output Bus (16-bits)
11001100	11111111 11001100
01010101	00000000 01010101



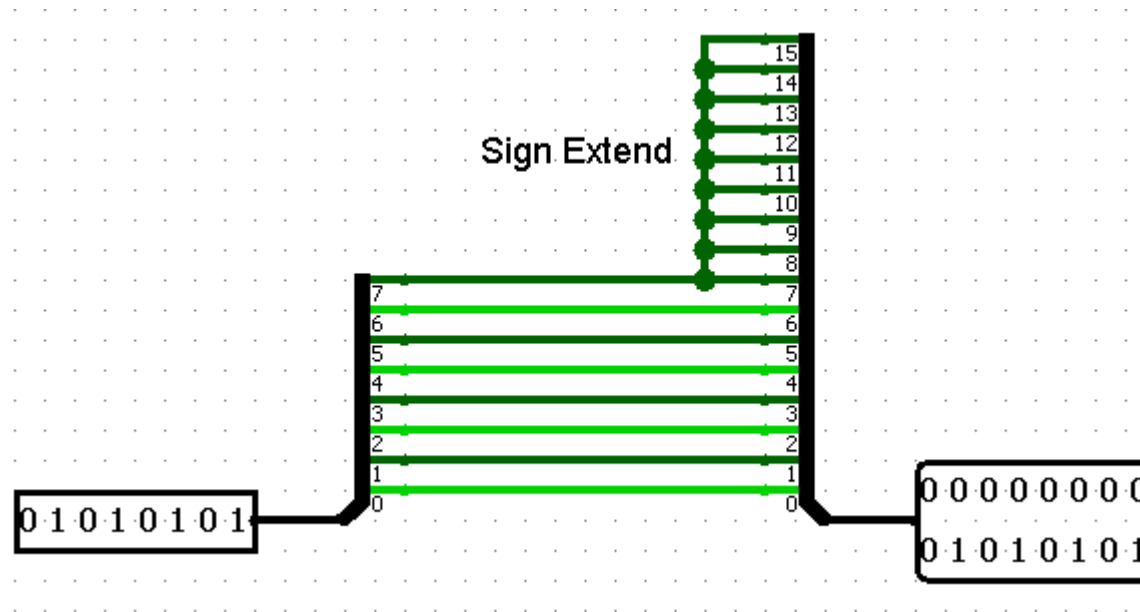
Sign Extend

Input Bus (8-bits)	Output Bus (16-bits)
11001100	11111111 11001100
01010101	00000000 01010101



Sign Extend

Input Bus (8-bits)	Output Bus (16-bits)
11001100	11111111 11001100
01010101	00000000 01010101



Encoders

- An encoder is a combinational logic circuit with 2^n input pins and n output pins.
 - Only a single input pin is set to 1

- Below is the truth table for a 4x2 encoder

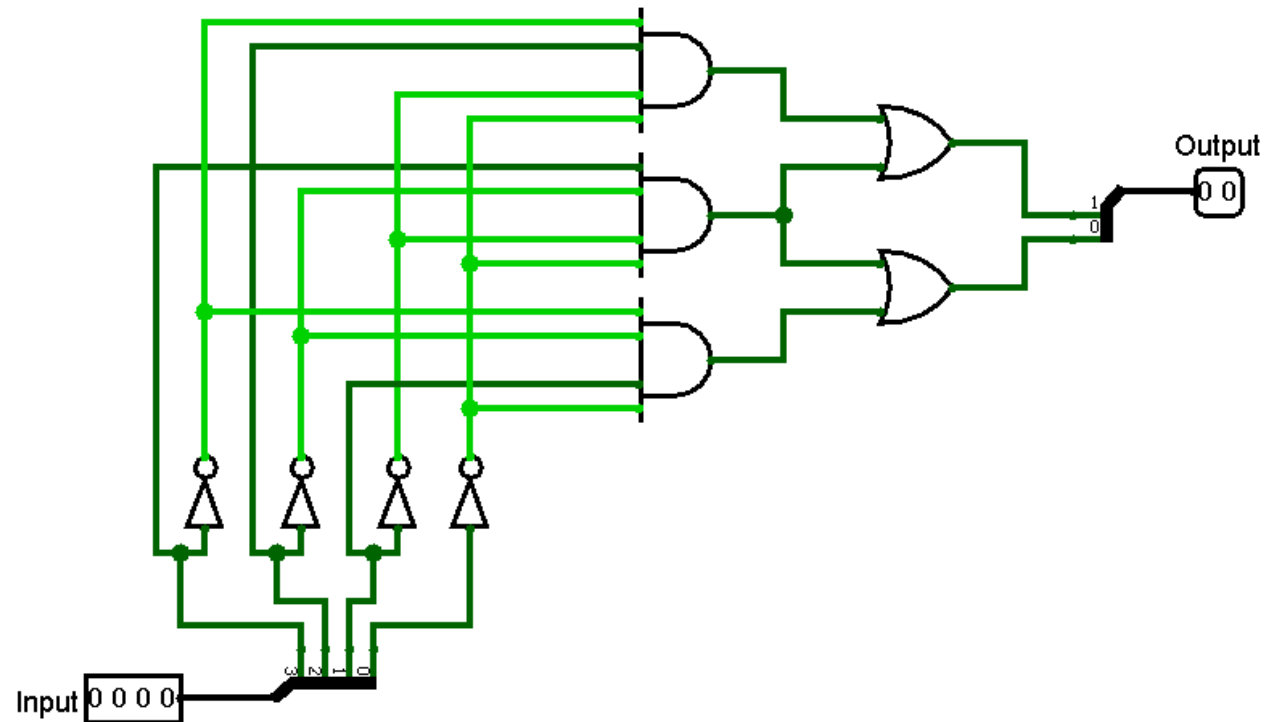
- I represents the input pins
- E represents the (encoded) output pins

I_3	I_2	I_1	I_0	E_1	E_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

- SOP Expression: $E_1 = \bar{I}_3 I_2 \bar{I}_1 \bar{I}_0 + I_3 \bar{I}_2 \bar{I}_1 \bar{I}_0$
 - SOP Expression: $E_0 = \bar{I}_3 \bar{I}_2 I_1 \bar{I}_0 + I_3 \bar{I}_2 \bar{I}_1 \bar{I}_0$
- Same term

Encoders

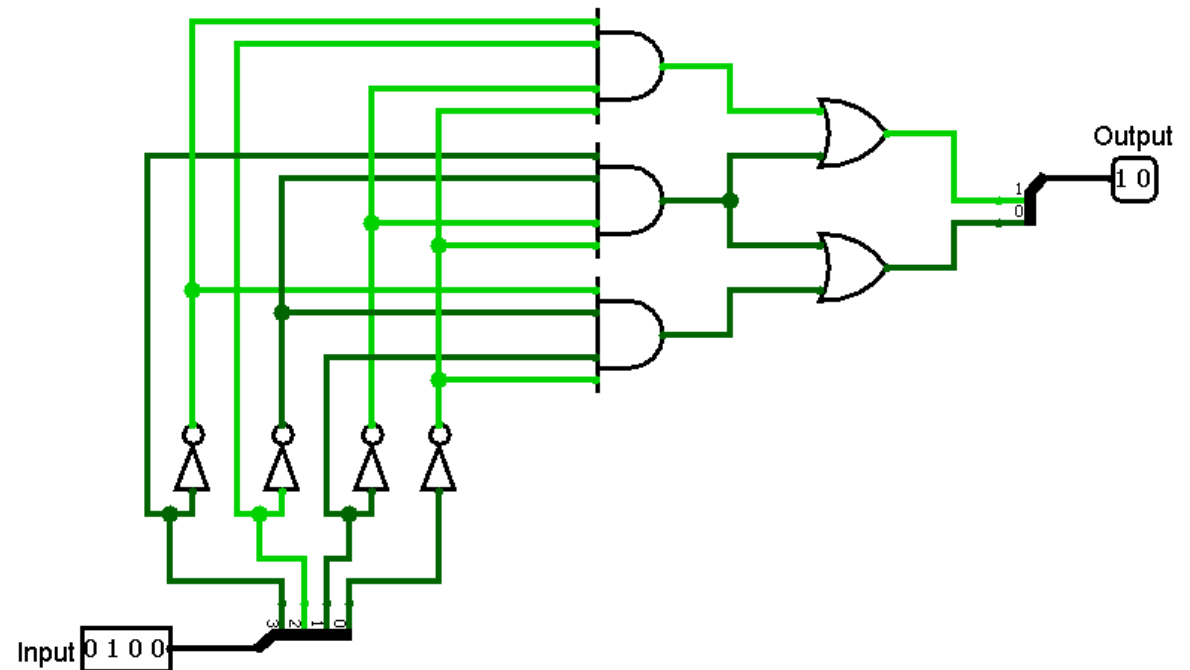
- Logic circuit for a 4x2 encoder:



Encoders

- Logic circuit for a 4x2 encoder:

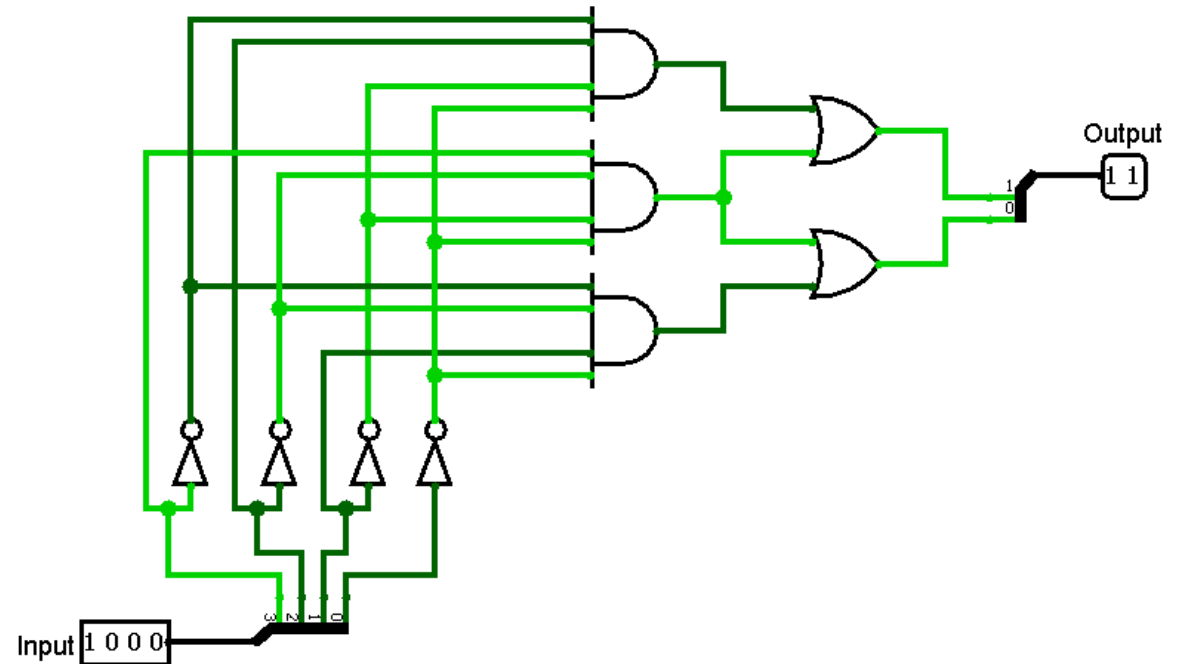
I_3	I_2	I_1	I_0	E_1	E_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



Encoders

- Logic circuit for a 4x2 encoder:

I_3	I_2	I_1	I_0	E_1	E_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



Encoders

- Below is the truth table for an 8x3 encoder

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	E_2	E_1	E_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Encoders

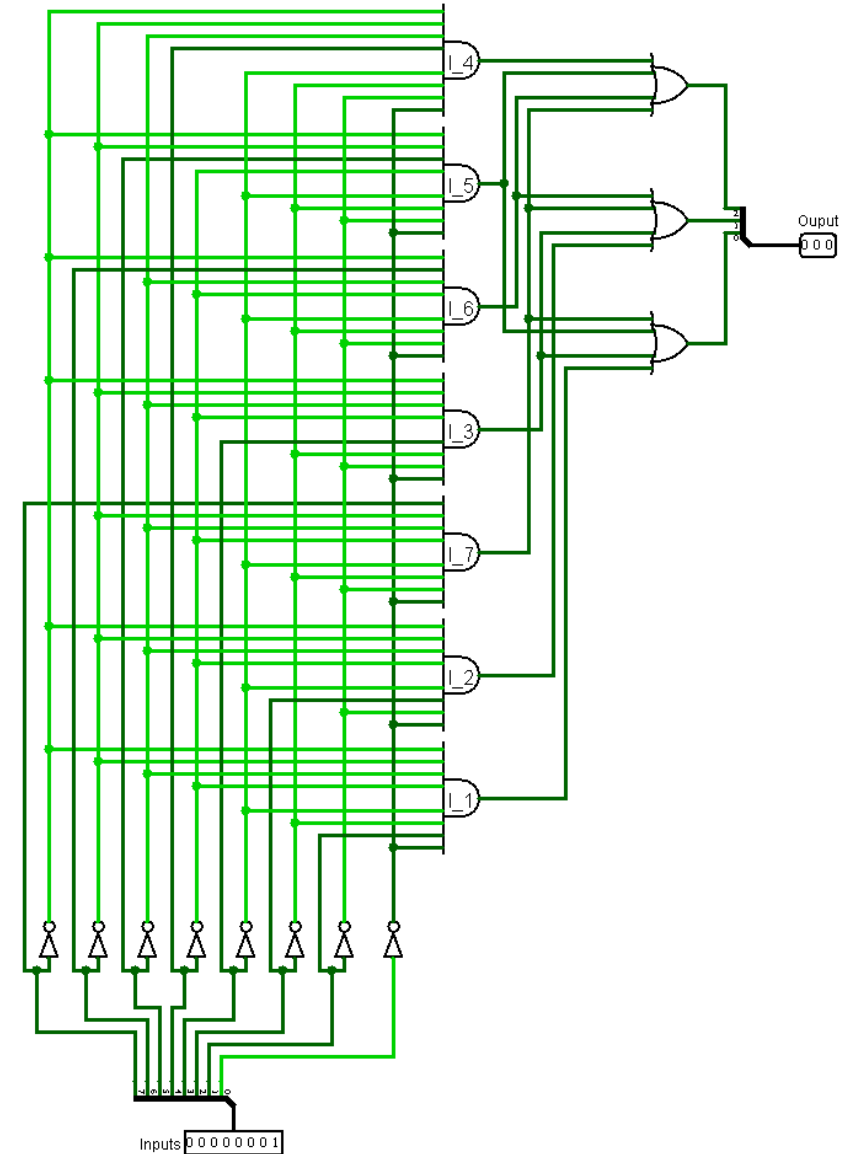
- SOP Expressions:

$$\begin{aligned} E_2 &= \bar{I}_7 \bar{I}_6 \bar{I}_5 I_4 \bar{I}_3 \bar{I}_2 \bar{I}_1 \bar{I}_0 + \boxed{\bar{I}_7 \bar{I}_6 I_5 \bar{I}_4 \bar{I}_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} + \boxed{\bar{I}_7 I_6 \bar{I}_5 \bar{I}_4 \bar{I}_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} + \boxed{I_7 \bar{I}_6 \bar{I}_5 \bar{I}_4 \bar{I}_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} \\ E_1 &= \bar{I}_7 \bar{I}_6 \bar{I}_5 \bar{I}_4 \bar{I}_3 I_2 \bar{I}_1 \bar{I}_0 + \boxed{\bar{I}_7 \bar{I}_6 \bar{I}_5 \bar{I}_4 I_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} + \boxed{\bar{I}_7 I_6 \bar{I}_5 \bar{I}_4 \bar{I}_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} + \boxed{I_7 \bar{I}_6 \bar{I}_5 \bar{I}_4 \bar{I}_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} \\ E_0 &= \bar{I}_7 \bar{I}_6 \bar{I}_5 \bar{I}_4 \bar{I}_3 \bar{I}_2 I_1 \bar{I}_0 + \boxed{\bar{I}_7 \bar{I}_6 \bar{I}_5 \bar{I}_4 I_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} + \boxed{\bar{I}_7 \bar{I}_6 I_5 \bar{I}_4 \bar{I}_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} + \boxed{I_7 \bar{I}_6 \bar{I}_5 \bar{I}_4 \bar{I}_3 \bar{I}_2 \bar{I}_1 \bar{I}_0} \end{aligned}$$

- (Boxed terms indicate duplicates)

Encoders

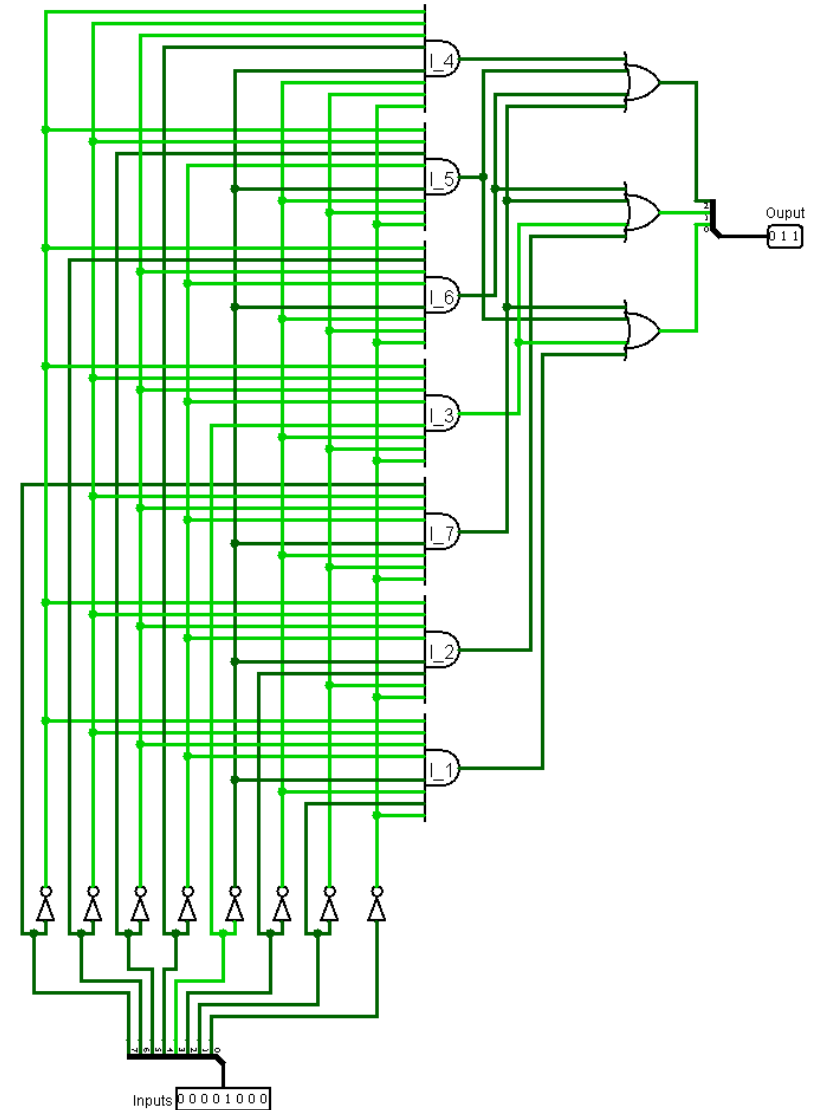
- Logic circuit for an 8x3 encoder:



Encoders

- Logic circuit for an 8x3 encoder:

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	E_2	E_1	E_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1



Decoders

- A decoder is a combinational logic circuit with n input pins and 2^n output pins.
 - For each output, only a single bit is set to 1

- Below is the truth table for a 2x4 decoder
 - I represents the input pins
 - D represents the (decoded) output pins

I_1	I_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

- SOP Expressions:

$$D_0 = \bar{I}_1 \bar{I}_0$$

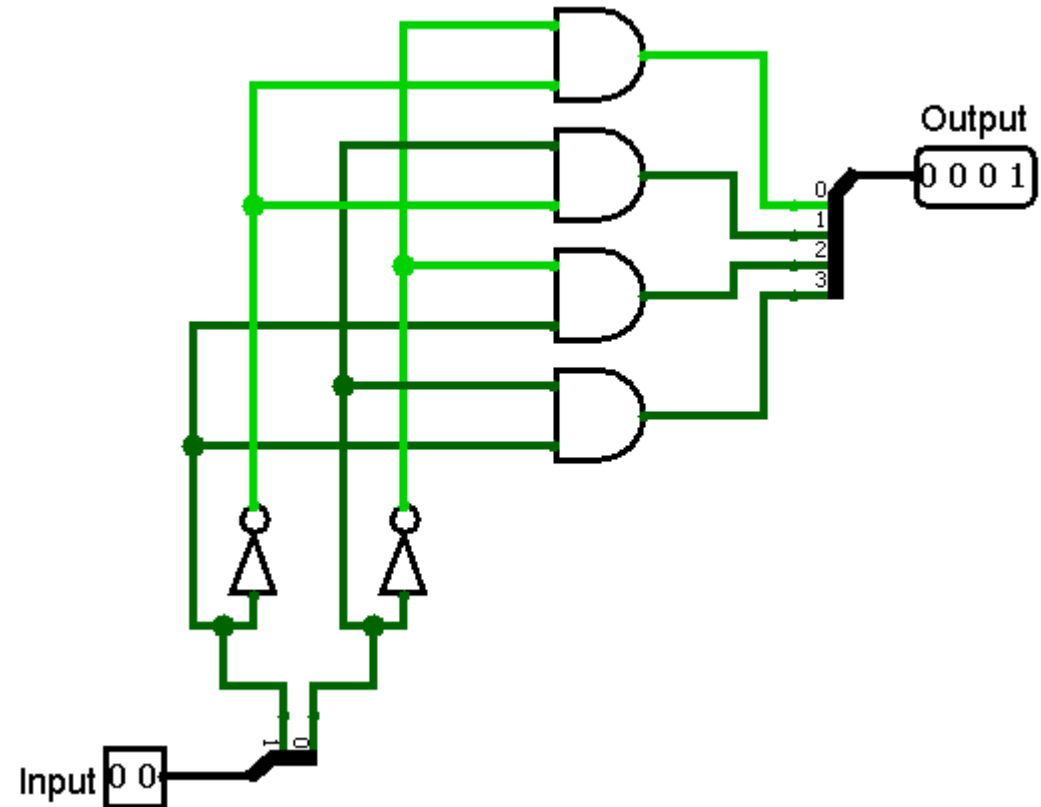
$$D_1 = \bar{I}_1 I_0$$

$$D_2 = I_1 \bar{I}_0$$

$$D_3 = I_1 I_0$$

Decoders

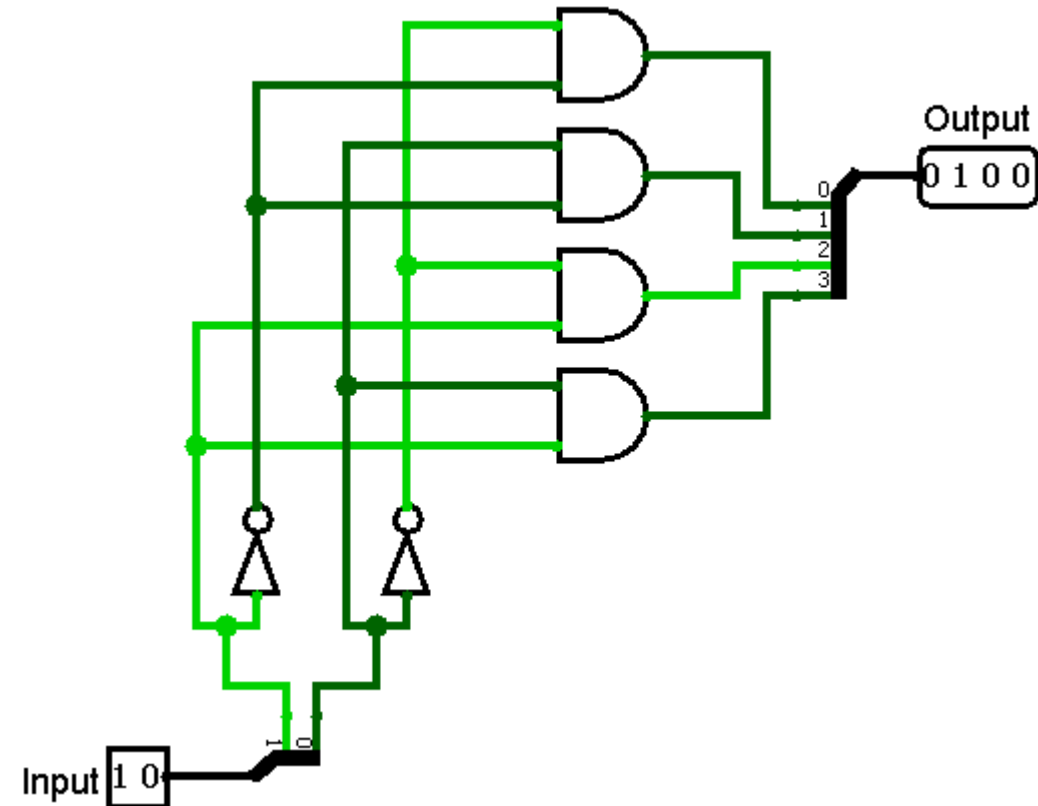
- Logic circuit for a 2x4 decoder



Decoders

- Logic circuit for a 2x4 decoder:

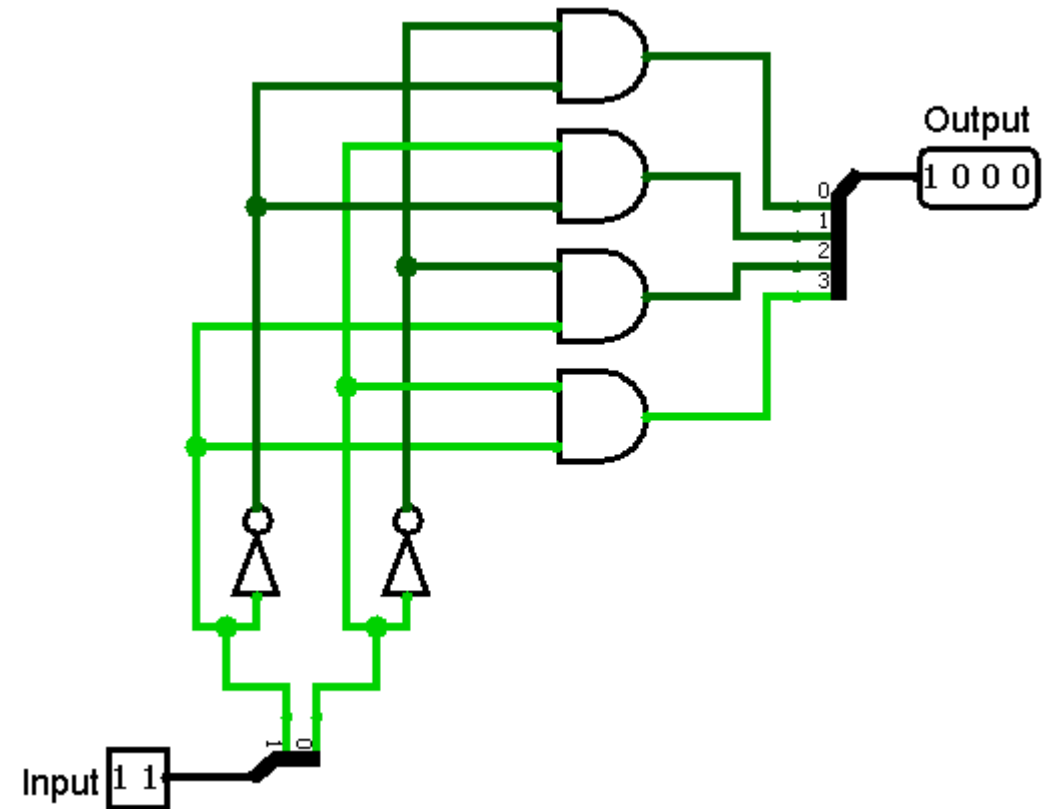
I_1	I_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



Decoders

- Logic circuit for a 2x4 decoder:

I_1	I_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



Decoders

- Below is the truth table for a 3x8 decoder

I_2	I_1	I_0	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

Decoders

- SOP Expressions:

$$D_0 = \bar{I}_2 \bar{I}_1 \bar{I}_0$$

$$D_1 = \bar{I}_2 \bar{I}_1 I_0$$

$$D_2 = \bar{I}_2 I_1 \bar{I}_0$$

$$D_3 = \bar{I}_2 I_1 I_0$$

$$D_4 = I_2 \bar{I}_1 \bar{I}_0$$

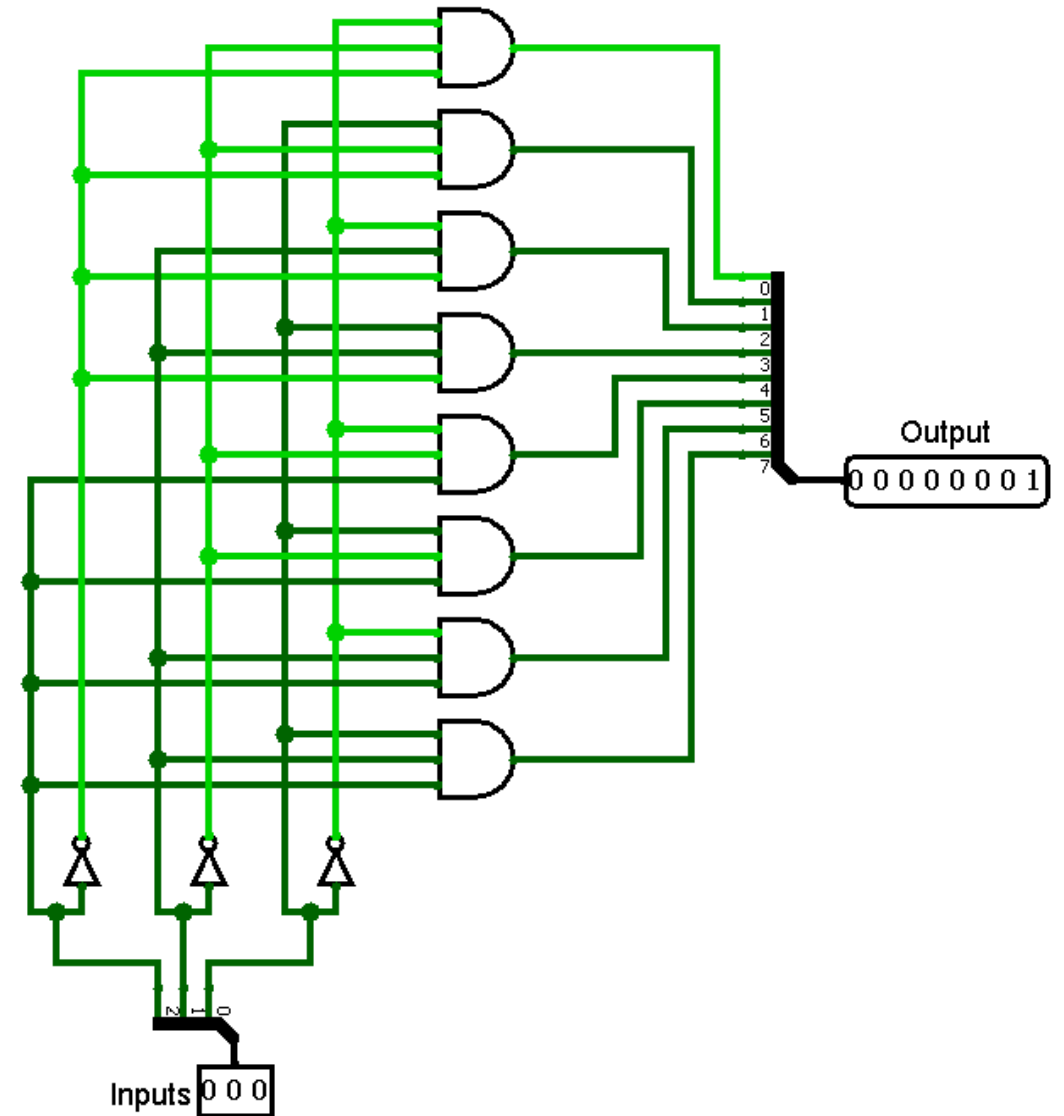
$$D_5 = I_2 \bar{I}_1 I_0$$

$$D_6 = I_2 I_1 \bar{I}_0$$

$$D_7 = I_2 I_1 I_0$$

Decoders

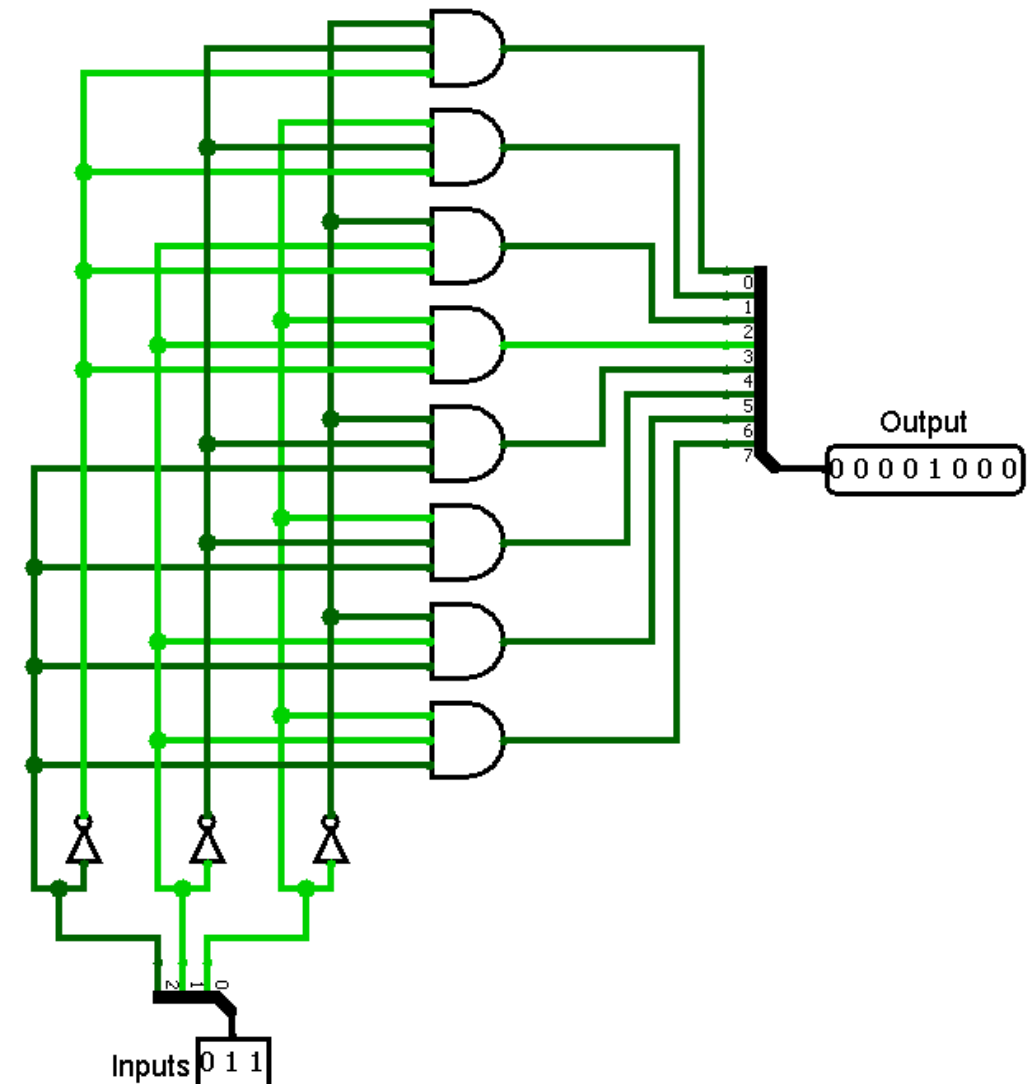
- Logic circuit for a 3x8 decoder:



Decoders

- Logic circuit for a 3x8 decoder:

I_2	I_1	I_0	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0



Multiplexers

- A multiplexer (*MUX*) is a combinational logic circuit used to select one of multiple input lines
- A multiplexer has 2^n input pins, n select pins, and 1 output pin.
 - The output is one of the input pins

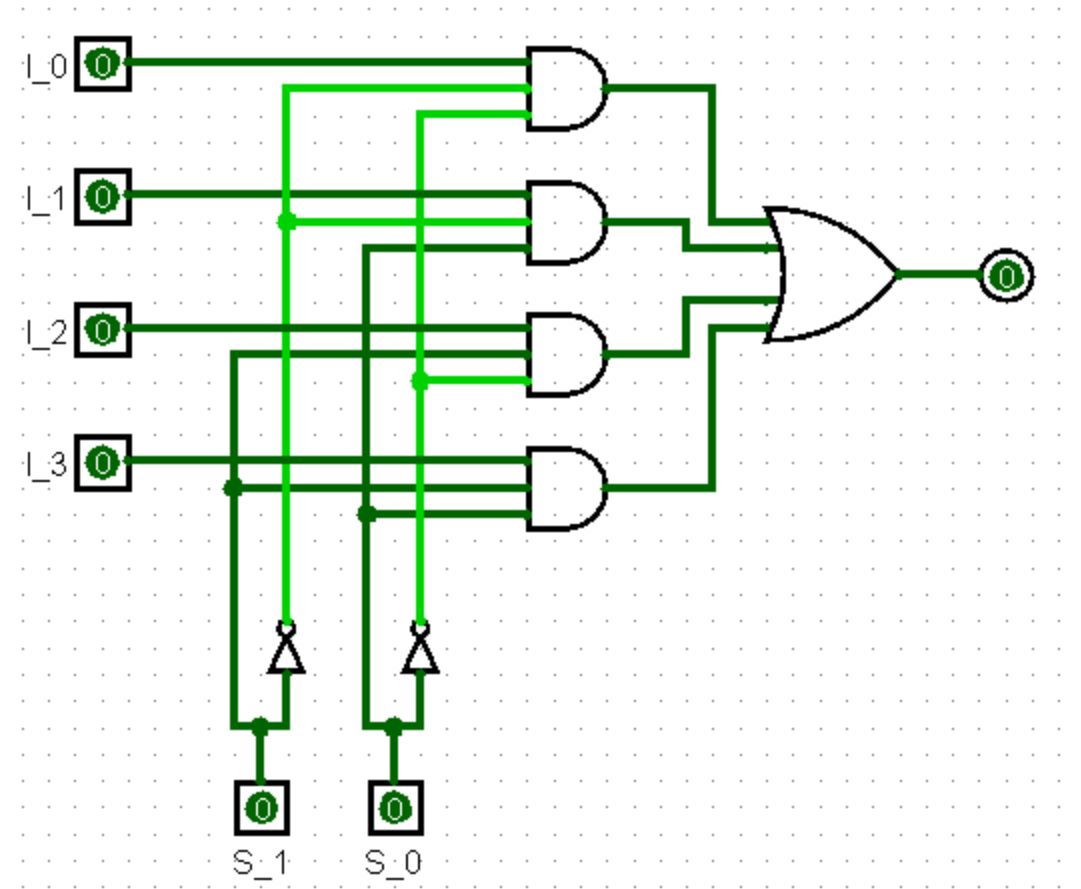
- The truth table for a 4x1 multiplexer:
 - *I* represents the input pins ($4 = 2^2$)
 - *S* represents the select pins (**2**)
 - *M* represents the output pin

S_1	S_0	M
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

SOP Expression: $M = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$

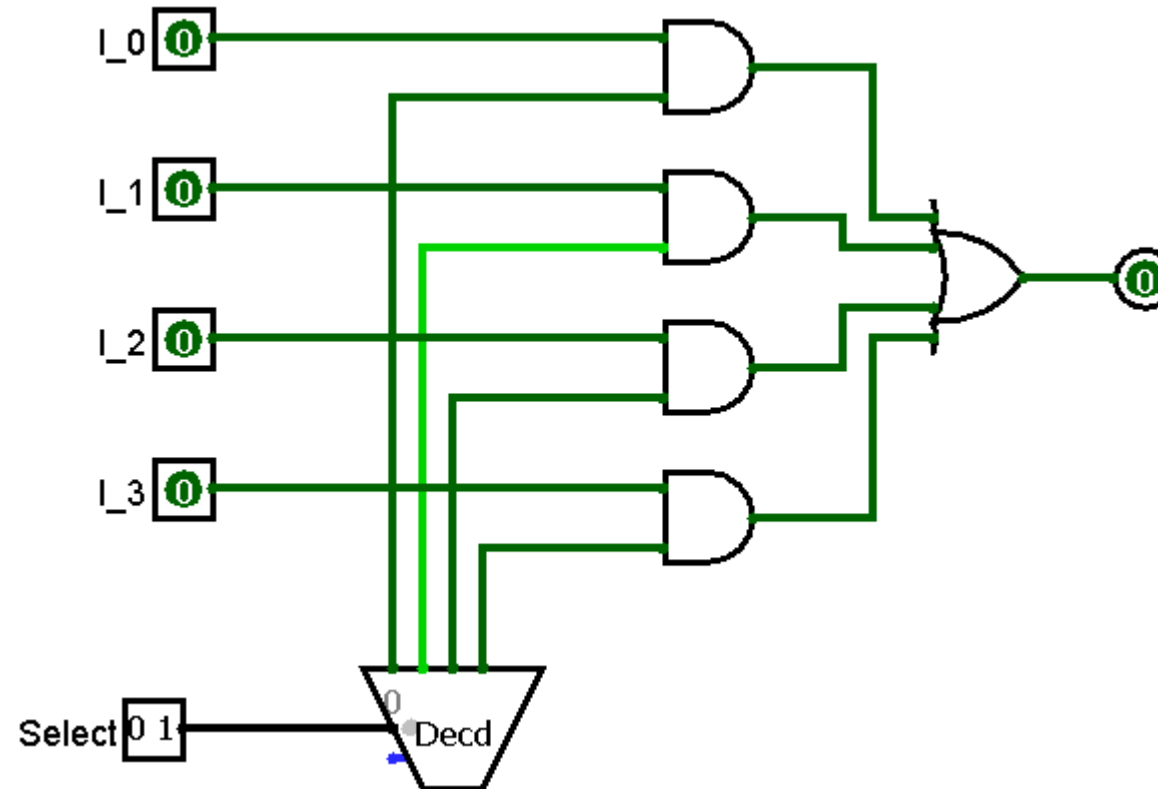
Multiplexers

- Logic circuit for a 4x1 multiplexer:



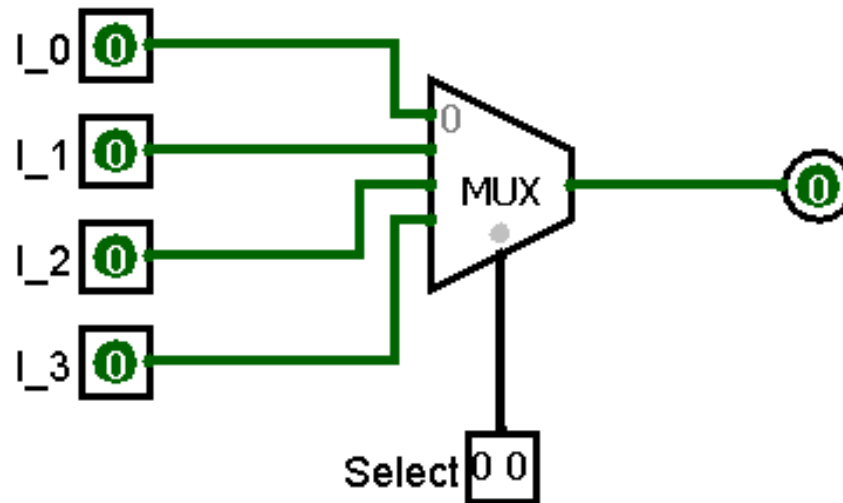
Multiplexers

- 4x1 multiplexer using a decoder:



Multiplexers

- Abstracted 4x1 multiplexer:



Multiplexers

- The truth table for an 8x1 multiplexer:
 - ***I*** represents the input pins ($8 = 2^3$)
 - ***S*** represents the select pins (**3**)
 - ***M*** represents the output pin

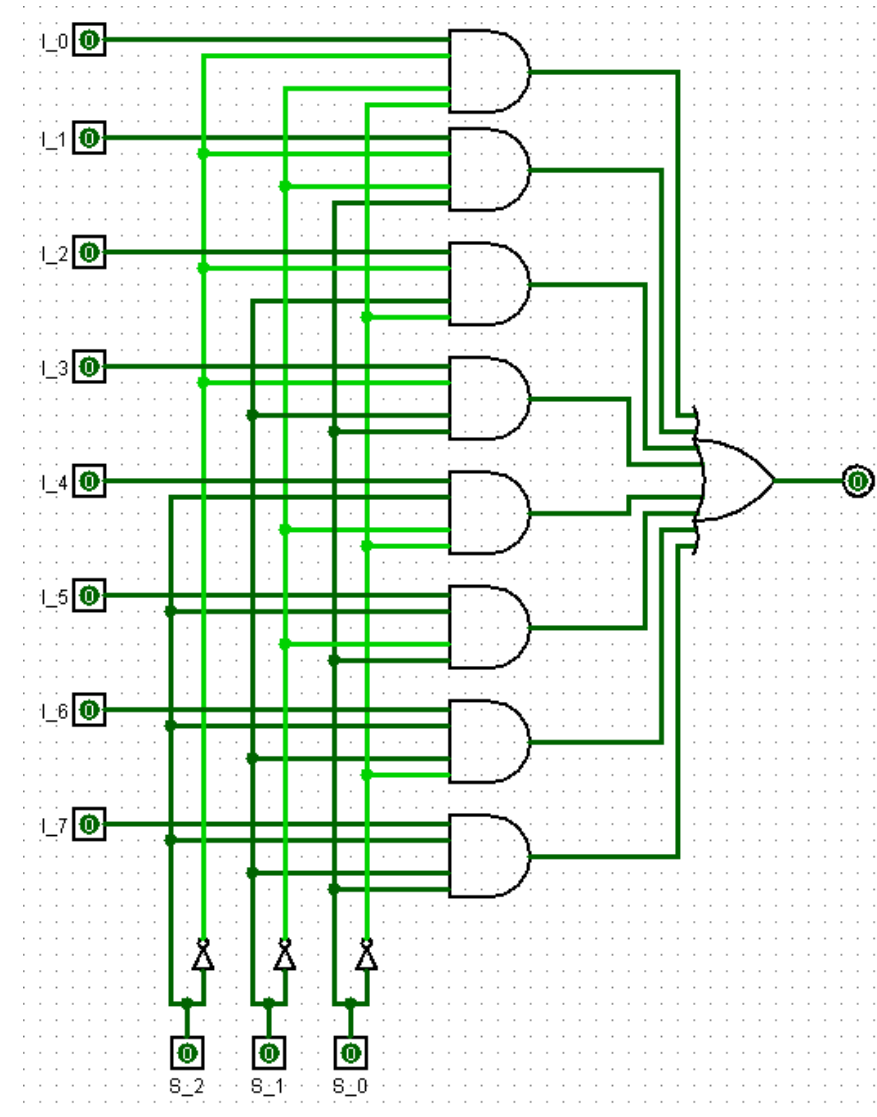
S_2	S_1	S_0	M
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

SOP Expression:

$$M = \bar{S}_2 \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_2 \bar{S}_1 S_0 I_1 + \bar{S}_2 S_1 \bar{S}_0 I_2 + \bar{S}_2 S_1 S_0 I_3 + S_2 \bar{S}_1 \bar{S}_0 I_4 + S_2 \bar{S}_1 S_0 I_5 + S_2 S_1 \bar{S}_0 I_6 + S_2 S_1 S_0 I_7$$

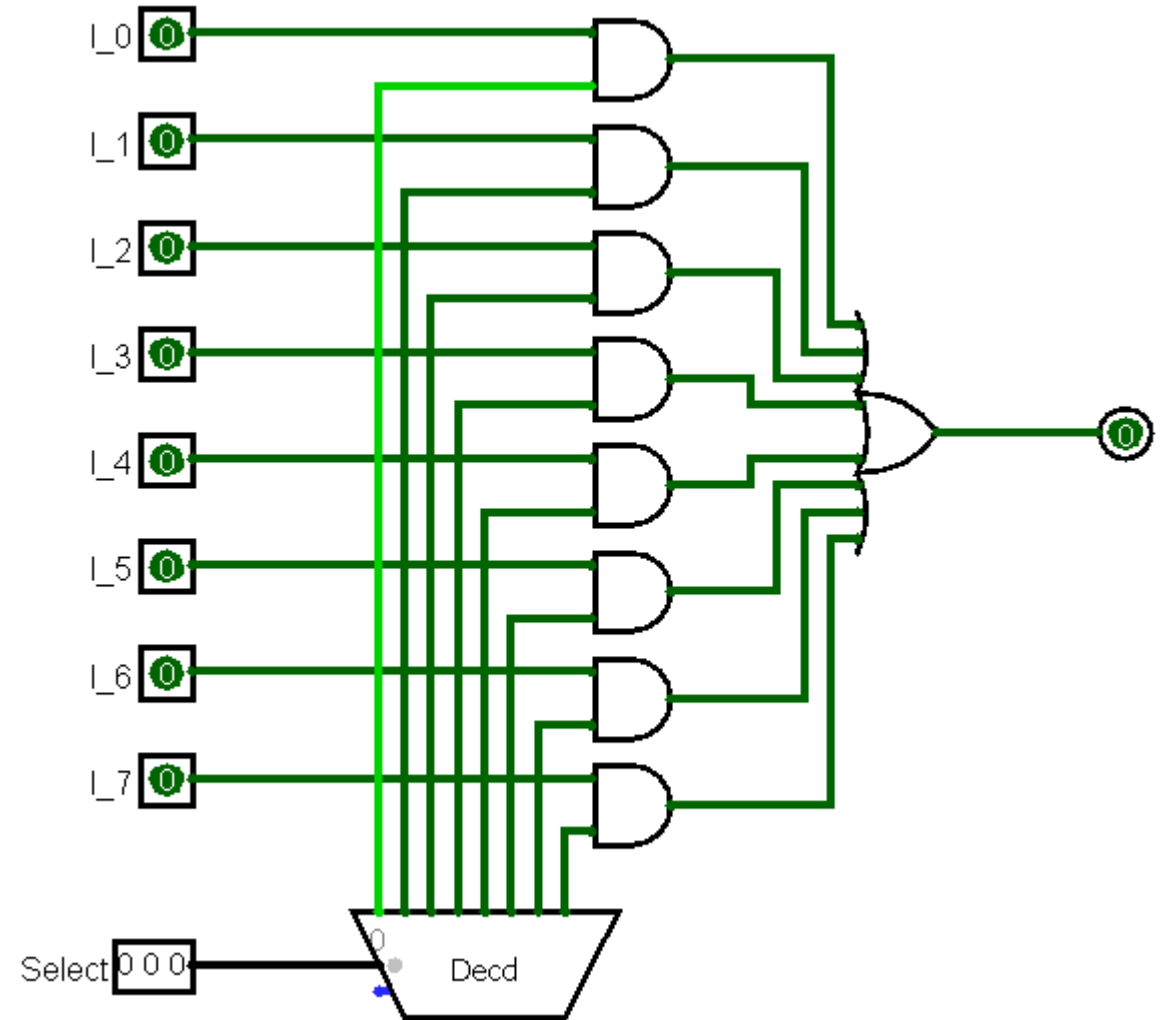
Multiplexers

- Logic circuit for an 8x1 multiplexer:



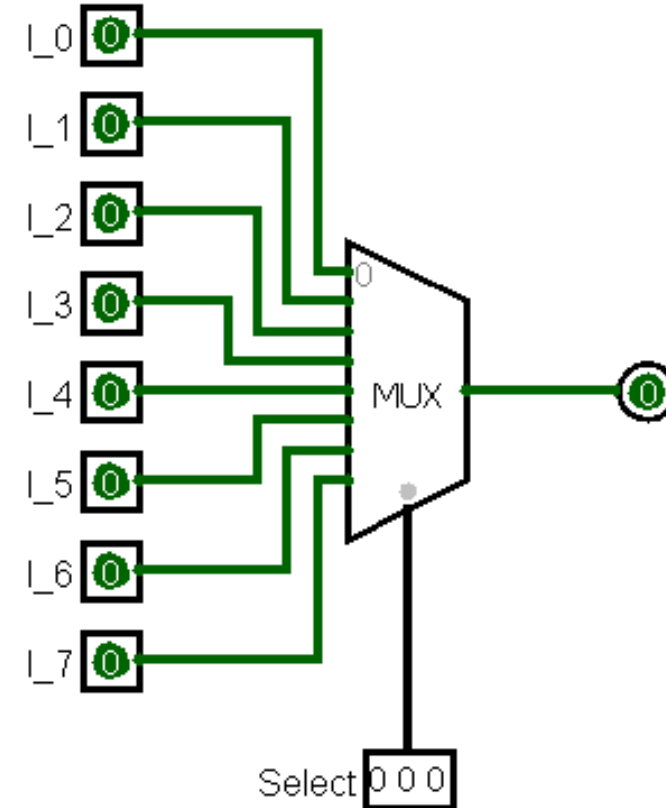
Multiplexers

- 8x1 multiplexer using a decoder:



Multiplexers

- Abstracted 8x1 multiplexer:



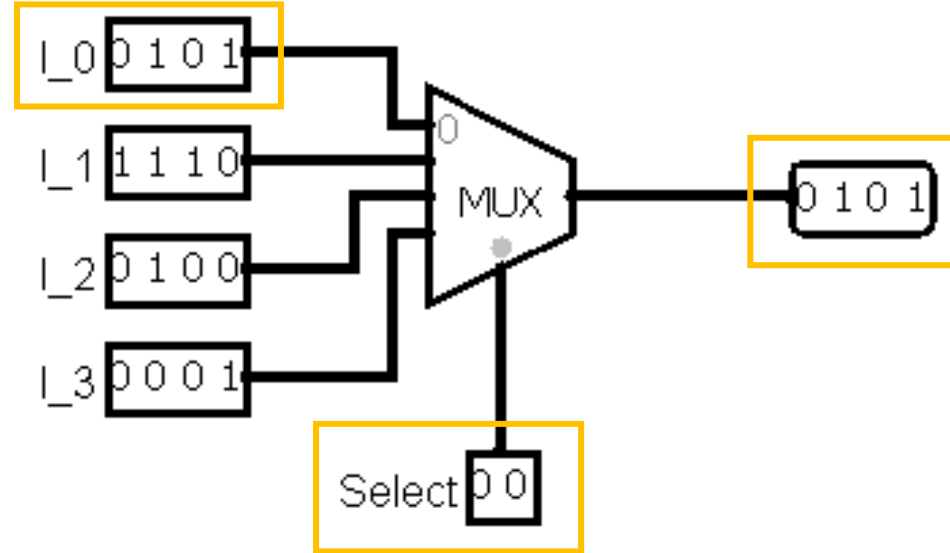
Multiplexers

- Buses can be inputs and outputs of a multiplexer.
 - A multiplexer with s select lines and n -bit buses is a $2^s \times n$ multiplexer
- For example, a multiplexer with 2 select lines (implying 4 inputs) where each input is a 4-bit bus
 - $2^s \times n = 2^2 \times 4 = 4 \times 4$ multiplexer

Multiplexers

- 4x4 Multiplexer:

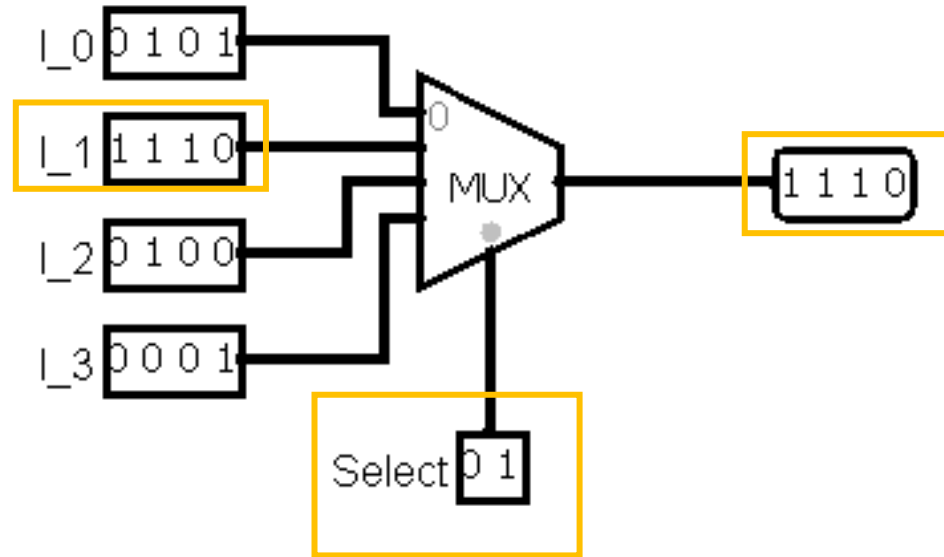
S_1	S_0	M
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



Multiplexers

- 4x4 Multiplexer:

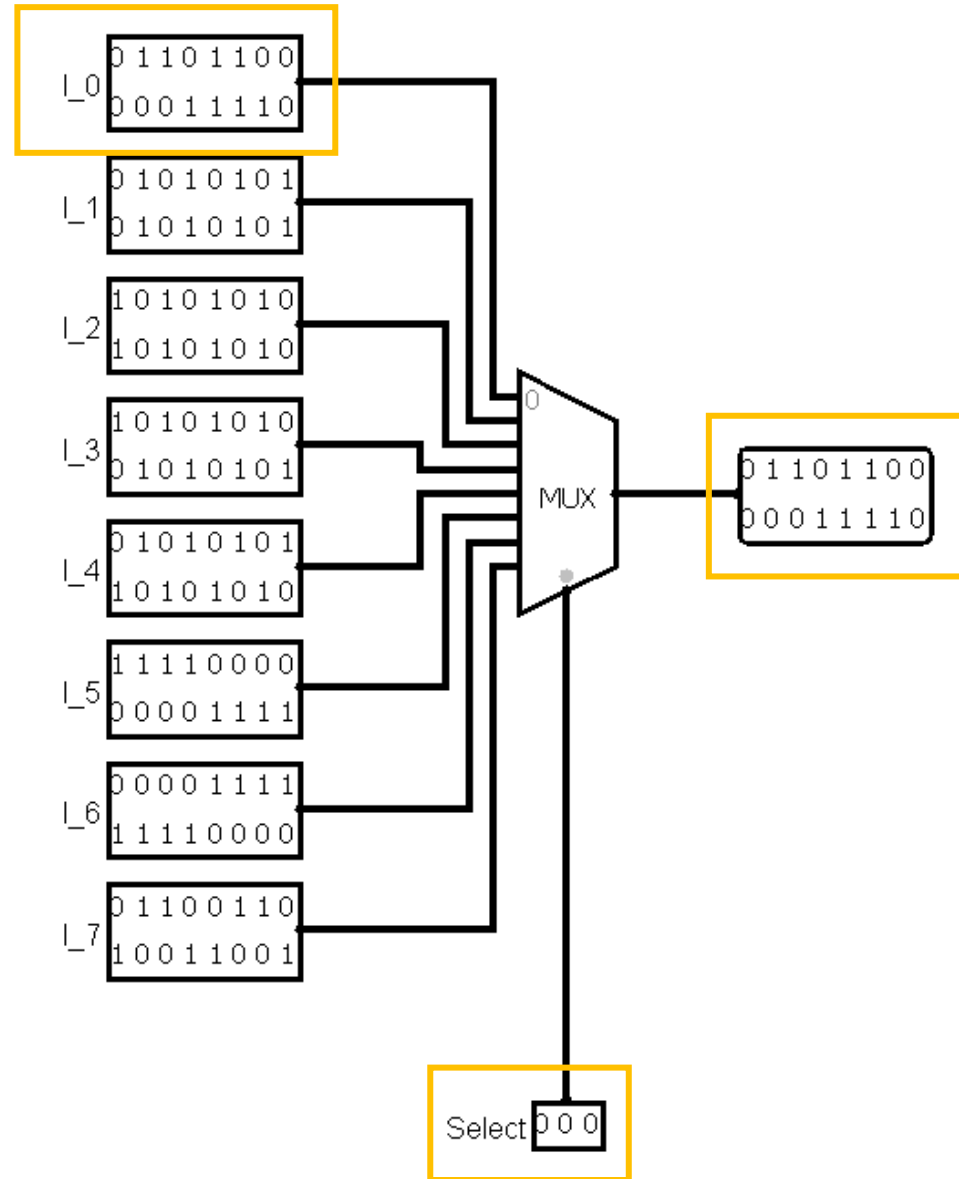
S_1	S_0	M
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



Multiplexers

- 8x16 Multiplexer:

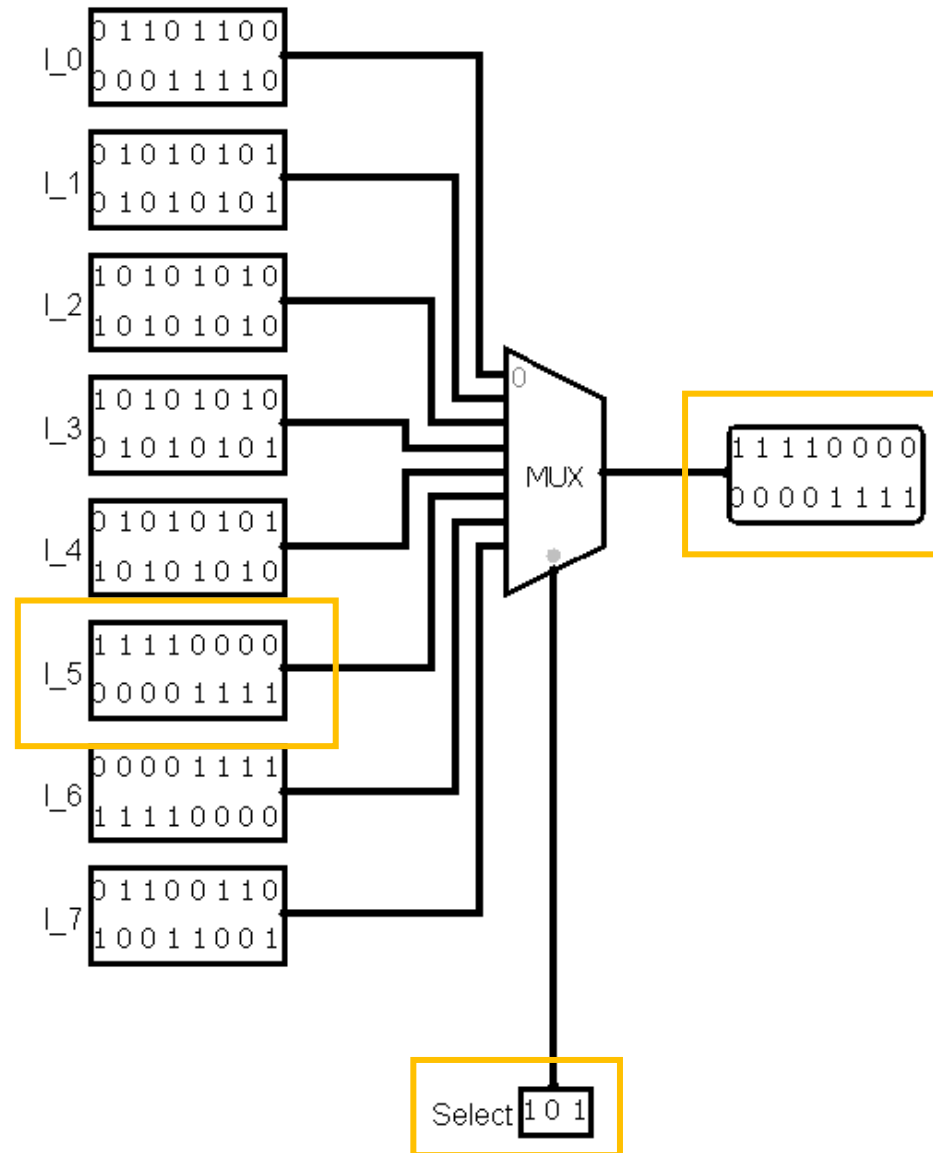
S_2	S_1	S_0	M
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7



Multiplexers

- 8x16 Multiplexer:

S_2	S_1	S_0	M
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7



Demultiplexers

- A demultiplexer (*DMX*) is a combinational logic circuit used to select the output from one input line
- A demultiplexer has 1 input pin, n select pins, and 2^n output pins.

- The truth table for a 1x4 demultiplexer:
 - *I* represents the input pin
 - *S* represents the select pins (2)
 - *D* represents the output pins ($4 = 2^2$)

<i>S</i> ₁	<i>S</i> ₀	<i>D</i> ₃	<i>D</i> ₂	<i>D</i> ₁	<i>D</i> ₀
0	0	0	0	0	<i>I</i>
0	1	0	0	<i>I</i>	0
1	0	0	<i>I</i>	0	0
1	1	<i>I</i>	0	0	0

Demultiplexers

SOP Expressions:

$$D_0 = \bar{S}_1 \bar{S}_0 I$$

$$D_1 = \bar{S}_1 S_0 I$$

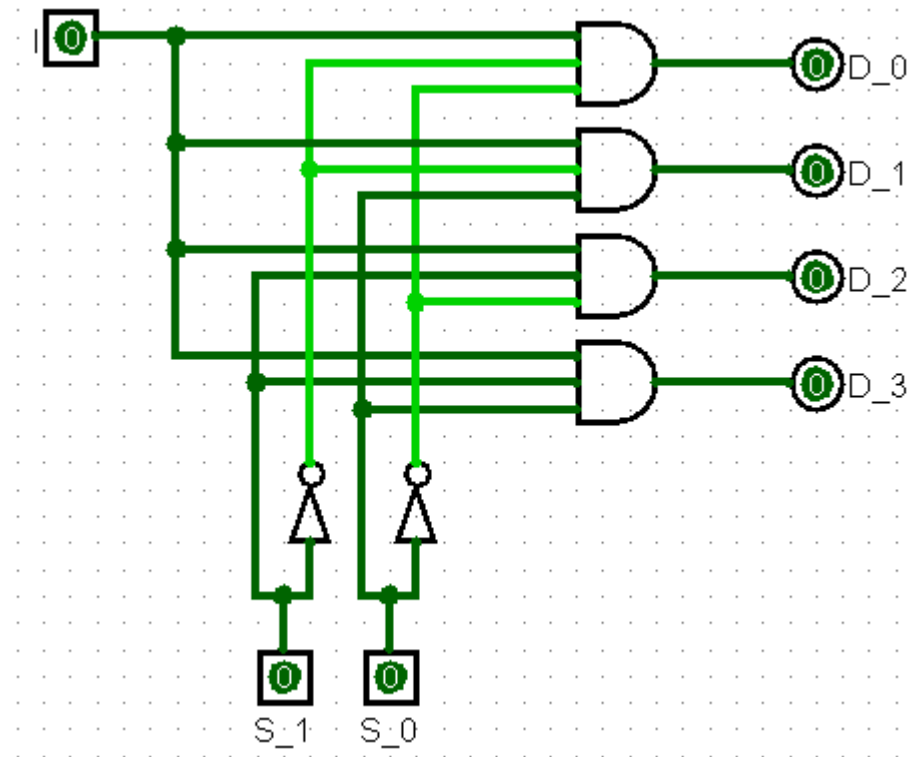
$$D_2 = S_1 \bar{S}_0 I$$

$$D_3 = S_1 S_0 I$$

S_1	S_0	D_3	D_2	D_1	D_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

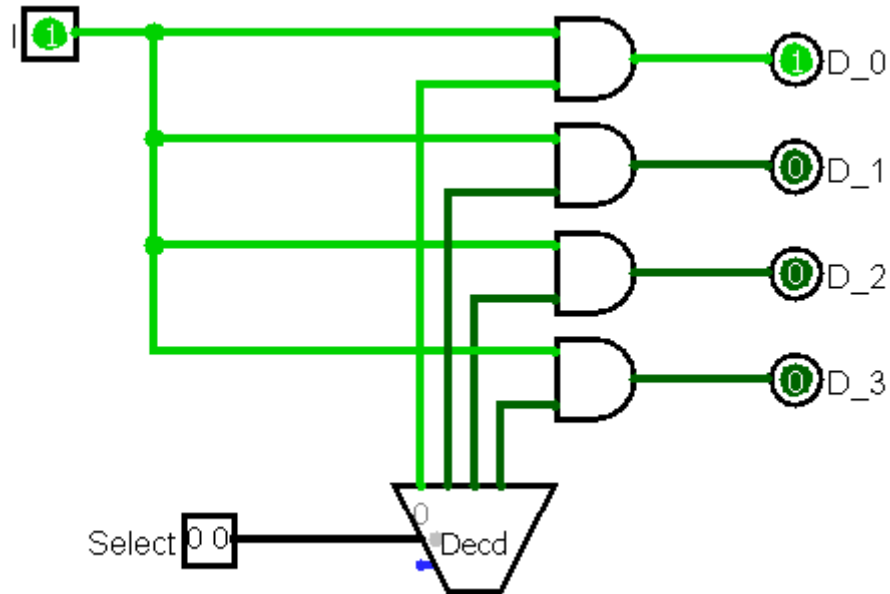
Demultiplexers

- 1x4 Demultiplexer:



Demultiplexers

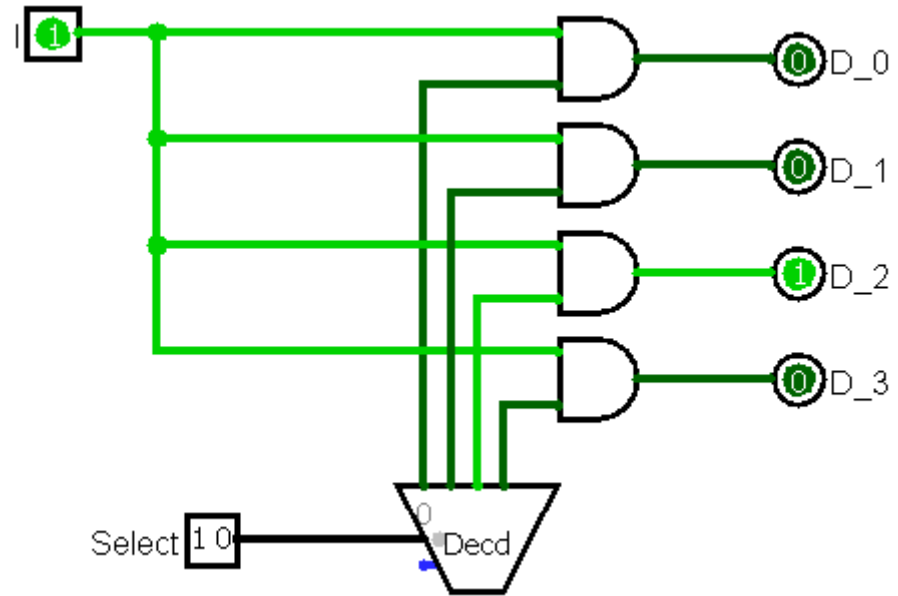
- 1x4 Demultiplexer using a decoder:



Demultiplexers

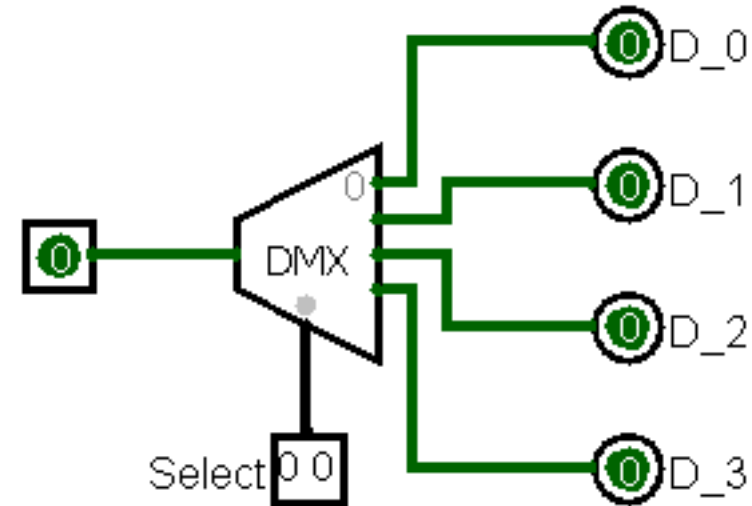
- 1x4 Demultiplexer:

S_1	S_0	D_3	D_2	D_1	D_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0



Demultiplexers

- Abstracted 1x4 demultiplexer:



Demultiplexers

- The truth table for a 1x8 multiplexer:
 - ***I*** represents the input pin
 - ***S*** represents the select pins (**3**)
 - ***D*** represents the output pins ($8 = 2^3$)

<i>S</i>₂	<i>S</i>₁	<i>S</i>₀	<i>D</i>₇	<i>D</i>₆	<i>D</i>₅	<i>D</i>₄	<i>D</i>₃	<i>D</i>₂	<i>D</i>₁	<i>D</i>₀
0	0	0	0	0	0	0	0	0	0	<i>I</i>
0	0	1	0	0	0	0	0	0	<i>I</i>	0
0	1	0	0	0	0	0	0	<i>I</i>	0	0
0	1	1	0	0	0	0	<i>I</i>	0	0	0
1	0	0	0	0	0	<i>I</i>	0	0	0	0
1	0	1	0	0	<i>I</i>	0	0	0	0	0
1	1	0	0	<i>I</i>	0	0	0	0	0	0
1	1	1	<i>I</i>	0	0	0	0	0	0	0

Demultiplexers

SOP Expressions:

$$D_0 = \overline{S_2} \overline{S_1} \overline{S_0} I$$

$$D_1 = \overline{S_2} \overline{S_1} S_0 I$$

$$D_2 = \overline{S_2} S_1 \overline{S_0} I$$

$$D_3 = \overline{S_2} S_1 S_0 I$$

$$D_4 = S_2 \overline{S_1} \overline{S_0} I$$

$$D_5 = S_2 \overline{S_1} S_0 I$$

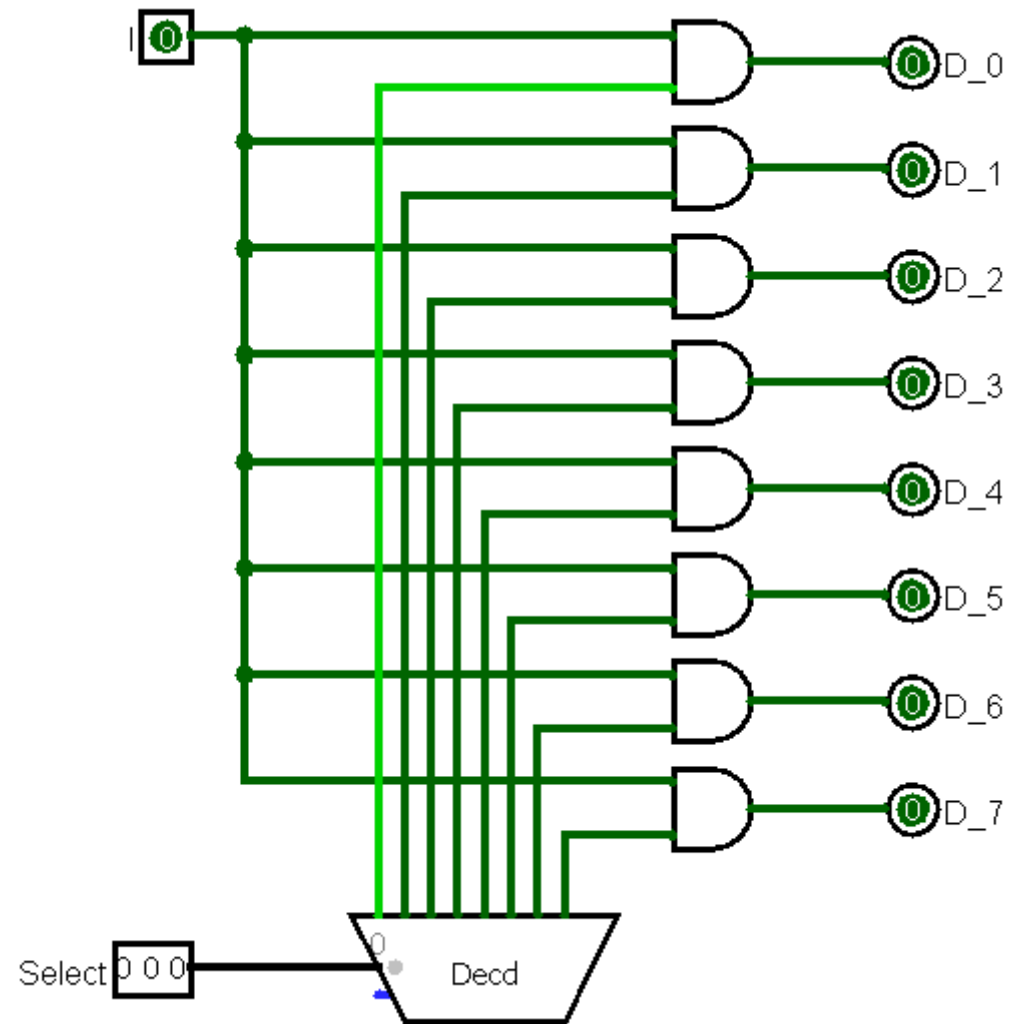
$$D_6 = S_2 S_1 \overline{S_0} I$$

$$D_7 = S_2 S_1 S_0 I$$

S_2	S_1	S_0	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0	<i>I</i>
0	0	1	0	0	0	0	0	0	<i>I</i>	0
0	1	0	0	0	0	0	0	<i>I</i>	0	0
0	1	1	0	0	0	0	<i>I</i>	0	0	0
1	0	0	0	0	0	<i>I</i>	0	0	0	0
1	0	1	0	0	<i>I</i>	0	0	0	0	0
1	1	0	0	<i>I</i>	0	0	0	0	0	0
1	1	1	<i>I</i>	0	0	0	0	0	0	0

Demultiplexers

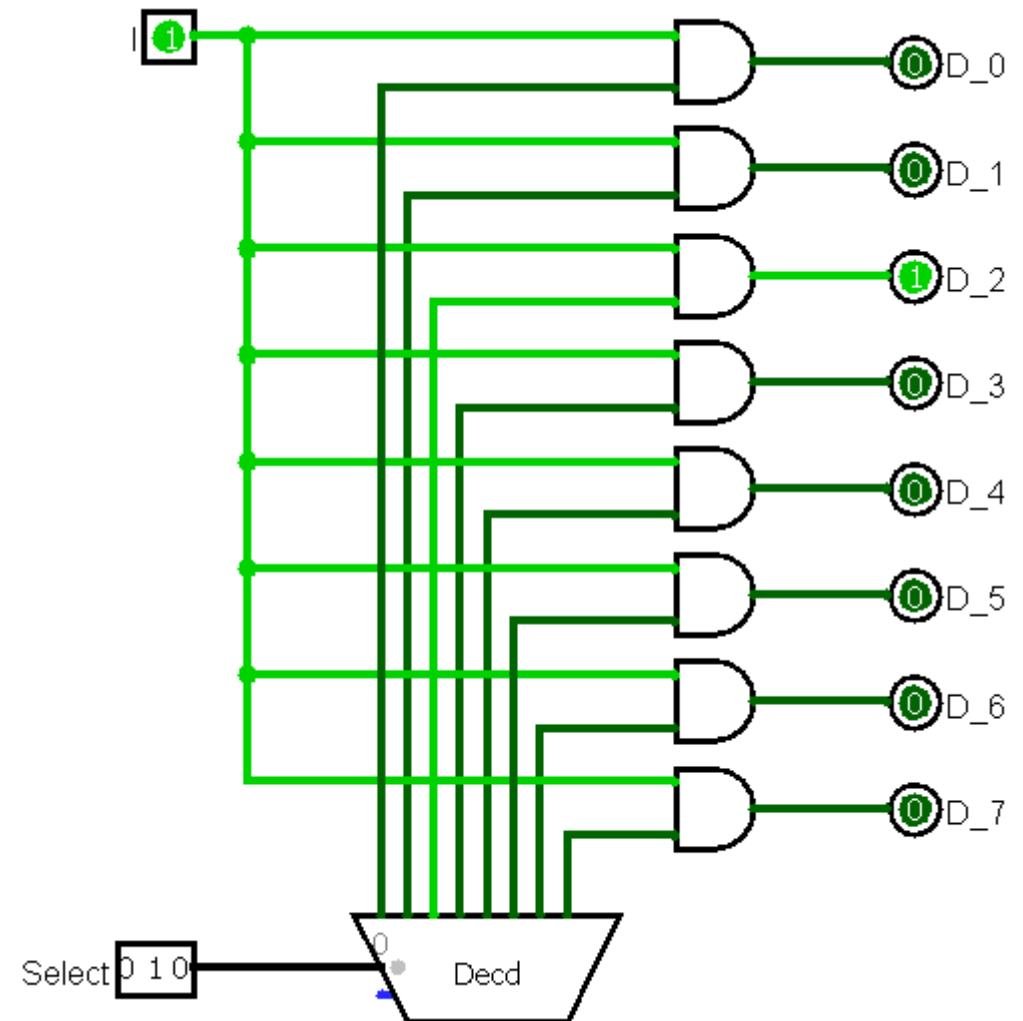
- 1x8 Demultiplexer:



Demultiplexers

- 1x8 Demultiplexer:

S_2	S_1	S_0	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0



Demultiplexers

- Abstracted 1x8 demultiplexer:

